

Finite Forward-Iterate Schröder Approximation from Geometric Richardson Extrapolation

Reciprocal Filters and Exact Paired Remainders

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Abstract

Let Φ be an odd analytic germ with $\Phi(0) = 0$ and $\Phi'(0) = 1$, let $H = \Phi^{-1}$, and fix $m > 1$. The conjugated multiplication map

$$S_m(x) = \Phi(mH(x))$$

satisfies the classical Schröder relation $H \circ S_m = mH$. For $y = H(x)$, consider the Richardson family

$$T_y(h) = \frac{\Phi(hy)}{h}.$$

At the geometric nodes $h = m^q$, one has the exact identity

$$T_y(m^q) = m^{-q} S_m^{\circ q}(x).$$

Thus geometric Richardson extrapolation of T_y to $h = 0$ becomes, in the original variable, a finite fixed-weight linear combination of forward iterates of the repelling map S_m . This note develops that realization and its reciprocal shrinking-scale form. The resulting composition and scale filters satisfy

$$\ell_r(z) = z^r p_r(z^{-1}), \quad p_r(z) = \prod_{j=1}^r \frac{m^{2j} - z}{m^{2j} - 1},$$

and give paired exact remainder identities. In the generic case

$$H(x) - P_{r+1}^{(m, \Phi)}(x) = (-1)^{r+1} a_{r+1} m^{r(r+1)} x^{2r+3} + O(x^{2r+5}),$$

where $\Phi(t) = t + \sum_{n \geq 1} a_n t^{2n+1}$. The Richardson filters, geometric-node error factors, skipped-mode order jumps, and the underlying Schröder–Koenigs linearization are classical inputs; the emphasis here is the explicit finite forward-iterate realization, its reciprocal pairing with the shrinking-scale filter, and the exact remainder formulas in the original variable.

1 Purpose, claim boundary, and related work

The construction in this note sits at the intersection of two classical theories. Schröder–Koenigs linearization converts a nonlinear map near a fixed point into multiplication in a conjugate coordinate,

while Richardson extrapolation removes prescribed asymptotic power modes by finite weighted combinations. Neither ingredient is claimed as new.

The observation developed here is deliberately narrower. If $H \circ S_m = mH$ and $y = H(x)$, then geometric Richardson data for

$$T_y(h) = \frac{\Phi(hy)}{h}$$

at the forward nodes $h = 1, m, \dots, m^r$ are exactly evaluations of finite forward iterates of S_m in the original variable. Consequently the Richardson extrapolant can be written without evaluating H :

$$P_{r+1}^{(m, \Phi)}(x) = \sum_{q=0}^r \lambda_{r,q} m^{-q} S_m^{\circ q}(x).$$

The corresponding shrinking-scale Richardson filter is the reciprocal polynomial $\ell_r(z) = z^r p_r(z^{-1})$. To the author's knowledge, this finite forward-iterate realization and its reciprocal pairing with the shrinking-scale filter have not previously been stated in this form. This is a literature-status statement, not an absolute priority claim.

Several parts of the calculation are standard and are used here as such. Schröder's equation, Koenigs linearization, and composition-operator eigenfunctions are classical [1, 2]. Targonski studied Schröder equations, iterated-function interpolation, and Chebyshev generalizations well before the present construction [3, 4, 5]. Mira's 1982 contribution on Schröder equations and generalized Chebyshev polynomials is closely adjacent to the map-construction side [6]; the full text was not available for direct inspection during this revision, so no priority claim concerning its detailed formulas is made here.

On the extrapolation side, geometric-node Richardson and its error analysis belong to the classical theory; see Joyce's survey, Brezinski–Redivo-Zaglia, and Sidi; see especially Sidi, Chapter 1, for the Richardson framework and the classical Archimedean π example [7, 8, 9]. In particular, vanishing asymptotic coefficients and the resulting order jumps are inherited Richardson phenomena rather than a separate novelty claim. The sine sequence $n \sin(1/n)$ also appears as a standard extrapolation example in the classical literature, so polygonal-sine or polygonal- π acceleration is treated below only as historical context and specialization.

Iserles developed constructive Schröder approximation and a substantial line of work on acceleration of functional iteration [10, 11, 12, 13]. In the 1993 paper the Schröder function is constructed through Taylor-coefficient recurrences and an auxiliary generating function. His iteration argument uses forward iterates of an attracting map, with $0 < |\beta| < 1$, either taken to the Koenigs limit or, in the acceleration work, supplied to nonlinear Hankel/Steffensen transforms. The construction here instead uses finitely many forward iterates of the repelling map S_m with fixed linear weights. The two routes are therefore related but distinct.

The goal of this note is consequently modest and concrete: to formulate that realization cleanly, derive exact paired remainders on the composition and scale sides, and separate the general analytic-germ statement from the special polynomial structure of the sine/Chebyshev case.

Throughout the main theorem, $m > 1$ is an arbitrary real number. Oddness of Φ restricts the spectral ladder to odd powers and the associated Richardson expansion to even powers. The underlying diagonalization does not itself require oddness. The restriction to odd integer m is needed only in the sine specialization when one wants S_m to be a polynomial.

2 Odd analytic conjugacy and the forward-iterate realization

Let

$$\Phi(t) = t + \sum_{n \geq 1} a_n t^{2n+1} \quad (1)$$

be an odd real-analytic germ at the origin, normalized by $\Phi'(0) = 1$. Let $H = \Phi^{-1}$ denote its local analytic inverse. For $m > 1$, define

$$S_m(x) = \Phi(mH(x)), \quad (A_m f)(x) = \frac{1}{m} f(S_m(x)). \quad (2)$$

Then

$$H(S_m(x)) = mH(x). \quad (3)$$

Lemma 2.1 (Diagonal action). *For every integer $k \geq 1$,*

$$A_m(H^k) = m^{k-1} H^k. \quad (4)$$

In particular,

$$A_m(H^{2n+1}) = m^{2n} H^{2n+1}, \quad n \geq 0. \quad (5)$$

Proof. Using (3),

$$A_m(H^k) = \frac{1}{m} H(S_m(x))^k = \frac{1}{m} (mH(x))^k = m^{k-1} H(x)^k.$$

□

2.1 Richardson realization at forward geometric nodes

Fix x sufficiently near the origin and write $y = H(x)$. Define

$$T_y(h) := \frac{\Phi(hy)}{h}. \quad (6)$$

By (1),

$$T_y(h) = y + \sum_{n \geq 1} a_n y^{2n+1} h^{2n}. \quad (7)$$

Thus the target value is $T_y(0) = y = H(x)$, understood by analytic continuation at $h = 0$. At the geometric nodes $h = m^q$,

$$\boxed{T_y(m^q) = m^{-q} \Phi(m^q H(x)) = m^{-q} S_m^{\circ q}(x) = A_m^q x.} \quad (8)$$

Choose a neighborhood U of the origin on which Φ is analytic and one-to-one, so that $H \circ \Phi = \text{id}$ on U . For a fixed stage r , the identity (8) is valid whenever $m^q H(x) \in U$ for every $0 \leq q \leq r$. This condition ensures both that all evaluations of Φ are defined and that the inverse relation is valid at every intermediate iterate. Under this condition the finite composition formula below is precisely geometric-node Richardson extrapolation of T_y to $h = 0$, written back in the original variable. The forward nodes grow away from zero; no limiting forward orbit is required.

2.2 Reciprocal filters

For $r \geq 0$, set

$$p_r(z) = \prod_{j=1}^r \frac{m^{2j} - z}{m^{2j} - 1}, \quad p_0(z) = 1, \quad (9)$$

and define the canonical finite iterate lift

$$P_{r+1}^{(m, \Phi)}(x) := p_r(A_m)x. \quad (10)$$

On the scale side define the reciprocal polynomial

$$\ell_r(z) = \prod_{j=1}^r \frac{m^{2j}z - 1}{m^{2j} - 1}. \quad (11)$$

Lemma 2.2 (Reciprocal filter identity). *For every $r \geq 0$,*

$$\boxed{\ell_r(z) = z^r p_r(z^{-1})}. \quad (12)$$

Proof. Directly,

$$z^r p_r(z^{-1}) = \prod_{j=1}^r \frac{z(m^{2j} - z^{-1})}{m^{2j} - 1} = \prod_{j=1}^r \frac{m^{2j}z - 1}{m^{2j} - 1}.$$

□

The identity (12) is the algebraic source of the composition/scale duality below.

3 Exact composition-side remainder

Because $x = \Phi(H)$, equation (1) becomes

$$x = H + \sum_{n \geq 1} a_n H^{2n+1}. \quad (13)$$

The filter p_r satisfies

$$p_r(1) = 1, \quad p_r(m^{2n}) = 0, \quad 1 \leq n \leq r. \quad (14)$$

Theorem 3.1 (Exact local active-mode remainder). *Let $R_\Phi \in (0, \infty]$ denote the radius of convergence of (1). For every $r \geq 0$, the canonical lift satisfies the convergent germ identity below; its displayed series converges whenever $m^r |H(x)| < R_\Phi$.*

$$\boxed{P_{r+1}^{(m, \Phi)} = H + \sum_{n \geq r+1} a_n p_r(m^{2n}) H^{2n+1}}. \quad (15)$$

Equivalently,

$$\boxed{H - P_{r+1}^{(m, \Phi)} = - \sum_{n \geq r+1} a_n p_r(m^{2n}) H^{2n+1}}. \quad (16)$$

Proof. Apply the finite operator polynomial $p_r(A_m)$ to (13). Lemma 2.1 gives

$$p_r(A_m)x = p_r(1)H + \sum_{n \geq 1} a_n p_r(m^{2n})H^{2n+1}.$$

Now use (14). □

Thus the usual leading defect is merely the first nonzero term in an exact filtered tail.

4 Finite iterate representation and exact intertwining

Write

$$p_r(z) = \sum_{q=0}^r \lambda_{r,q} z^q. \quad (17)$$

Since

$$A_m^q x = m^{-q} S_m^{\circ q}(x), \quad (18)$$

we have the finite iterate representation

$$\boxed{P_{r+1}^{(m,\Phi)}(x) = \sum_{q=0}^r \lambda_{r,q} m^{-q} S_m^{\circ q}(x).} \quad (19)$$

Fix a real target Λ . Choose the univalence neighborhood U as above and take $N > 0$ sufficiently large that $m^q \Lambda / (m^r N) \in U$ for every $0 \leq q \leq r$; equivalently, all intermediate points required by the orbit identity remain in the chosen Schröder coordinate chart. Then define

$$G_N^{(\Phi,\Lambda)} = N \Phi\left(\frac{\Lambda}{N}\right). \quad (20)$$

Let

$$(\mathcal{D}_m G)_N = G_{mN}, \quad \mathcal{L}_r^{(m)} := \ell_r(\mathcal{D}_m) = \prod_{j=1}^r \frac{m^{2j} \mathcal{D}_m - I}{m^{2j} - 1}. \quad (21)$$

When a genuinely discrete sequence is desired, one may restrict to integer $m \geq 2$ and integer N ; no such arithmetic restriction is needed for the analytic identity itself.

Set

$$x_K := \Phi\left(\frac{\Lambda}{K}\right). \quad (22)$$

Then

$$S_m(x_{mK}) = x_K, \quad S_m^{\circ q}(x_{m^r N}) = x_{m^{r-q} N}. \quad (23)$$

Theorem 4.1 (Exact Schröder–Richardson intertwining). *For every $r \geq 0$ and every N satisfying the preceding coordinate-domain condition,*

$$\boxed{\mathcal{L}_r^{(m)} G_N^{(\Phi,\Lambda)} = m^r N P_{r+1}^{(m,\Phi)}\left(\Phi\left(\frac{\Lambda}{m^r N}\right)\right).} \quad (24)$$

Consequently,

$$\boxed{\Lambda - \mathcal{L}_r^{(m)} G_N^{(\Phi,\Lambda)} = m^r N \left[H(x_{m^r N}) - P_{r+1}^{(m,\Phi)}(x_{m^r N}) \right].} \quad (25)$$

Proof. Using (19) and (23),

$$\begin{aligned} m^r N P_{r+1}^{(m, \Phi)}(x_{m^r N}) &= \sum_{q=0}^r \lambda_{r,q} m^{r-q} N x_{m^{r-q} N} \\ &= \sum_{q=0}^r \lambda_{r,q} G_{m^{r-q} N}^{(\Phi, \Lambda)}. \end{aligned}$$

On the other hand, Lemma 2.2 and (17) give

$$\ell_r(z) = z^r p_r(z^{-1}) = \sum_{q=0}^r \lambda_{r,q} z^{r-q}.$$

Therefore

$$\mathcal{L}_r^{(m)} G_N^{(\Phi, \Lambda)} = \sum_{q=0}^r \lambda_{r,q} G_{m^{r-q} N}^{(\Phi, \Lambda)},$$

which proves (24). Since

$$H(x_{m^r N}) = \frac{\Lambda}{m^r N},$$

subtracting (24) from Λ gives (25). $\square$

The reciprocal identity $\ell_r(z) = z^r p_r(z^{-1})$ therefore converts the composition-side filter into the scale-side filter exactly.

5 Exact sequence remainder

The scale expansion corresponding to (13) is

$$G_N^{(\Phi, \Lambda)} = \Lambda + \sum_{n \geq 1} a_n \Lambda^{2n+1} N^{-2n}. \quad (26)$$

Lemma 5.1 (Scale eigenvalues). *For every $n \geq 1$,*

$$\boxed{\mathcal{L}_r^{(m)}(N^{-2n}) = m^{-2rn} p_r(m^{2n}) N^{-2n}.} \quad (27)$$

Proof. Since $\mathcal{D}_m(N^{-2n}) = m^{-2n} N^{-2n}$, the multiplier contributed by $\mathcal{L}_r^{(m)}$ is

$$\prod_{j=1}^r \frac{m^{2j-2n} - 1}{m^{2j} - 1}.$$

For each factor,

$$m^{2j-2n} - 1 = m^{-2n} (m^{2j} - m^{2n}).$$

Extracting m^{-2n} from all r factors gives (27). $\square$

Theorem 5.2 (Exact scale-side active-mode remainder). *For all sufficiently large N ,*

$$\boxed{\Lambda - \mathcal{L}_r^{(m)} G_N^{(\Phi, \Lambda)} = - \sum_{n \geq r+1} a_n p_r(m^{2n}) m^{-2rn} \Lambda^{2n+1} N^{-2n}.} \quad (28)$$

Proof. Apply $\mathcal{L}_r^{(m)}$ termwise to the convergent expansion (26). The constant mode is preserved, while Lemma 5.1 gives the multiplier of each inverse-power mode. For $1 \leq n \leq r$, the factor $p_r(m^{2n})$ vanishes. $\square$

Equations (16) and (28) display the same active coefficients and the same spectral factor $p_r(m^{2n})$; only the reciprocal scale multiplier m^{-2rn} separates the two sides.

6 Skipped modes and exact first-surviving coefficients

The order-jump mechanism in this section is the familiar Richardson phenomenon that occurs when one or more nominal asymptotic coefficients vanish. The purpose here is not to claim a new superconvergence principle, but to record its exact specialization to the present conjugacy and the closed coefficient inherited from the geometric filter.

Assume the filtered tail is not identically zero, and define its first active index

$$n_\star := \min\{n \geq r+1 : a_n \neq 0\}. \quad (29)$$

For $q > 0$, define the Gaussian binomial coefficient by

$$\begin{bmatrix} L \\ R \end{bmatrix}_q := \prod_{j=1}^R \frac{q^{L+1-j} - 1}{q^j - 1}, \quad 0 \leq R \leq L. \quad (30)$$

Lemma 6.1 (First-surviving filter value). *For every $n_\star \geq r+1$,*

$$p_r(m^{2n_\star}) = (-1)^r m^{r(r+1)} \begin{bmatrix} n_\star - 1 \\ r \end{bmatrix}_{m^2}. \quad (31)$$

Proof. Starting from (9),

$$\begin{aligned} p_r(m^{2n_\star}) &= \prod_{j=1}^r \frac{m^{2j} - m^{2n_\star}}{m^{2j} - 1} \\ &= (-1)^r m^{2 \sum_{j=1}^r j} \prod_{j=1}^r \frac{m^{2(n_\star-j)} - 1}{m^{2j} - 1}. \end{aligned}$$

Since $2 \sum_{j=1}^r j = r(r+1)$, setting $q = m^2$ turns the remaining product into $\begin{bmatrix} n_\star - 1 \\ r \end{bmatrix}_q$. $\square$

Theorem 6.2 (Exact first-surviving coefficient). *Let $n_\star$ be given by (29). Then*

$$H(x) - P_{r+1}^{(m, \Phi)}(x) = (-1)^{r+1} a_{n_\star} m^{r(r+1)} \begin{bmatrix} n_\star - 1 \\ r \end{bmatrix}_{m^2} x^{2n_\star+1} + O(x^{2n_\star+3}). \quad (32)$$

On the scale side,

$$\Lambda - \mathcal{L}_r^{(m)} G_N^{(\Phi, \Lambda)} = (-1)^{r+1} a_{n_\star} \begin{bmatrix} n_\star - 1 \\ r \end{bmatrix}_{m^2} m^{-r(2n_\star-r-1)} \Lambda^{2n_\star+1} N^{-2n_\star} + O(N^{-2n_\star-2}). \quad (33)$$

Proof. In (16), the first surviving term is $-a_{n_*}p_r(m^{2n_*})H^{2n_*+1}$. Since $H(x) = x + O(x^3)$,

$$H^{2n_*+1} = x^{2n_*+1} + O(x^{2n_*+3}).$$

Substitution of (31) proves (32).

Likewise, the first surviving term of (28) is

$$-a_{n_*}p_r(m^{2n_*})m^{-2rn_*}\Lambda^{2n_*+1}N^{-2n_*}.$$

Using (31) and

$$r(r+1) - 2rn_* = -r(2n_* - r - 1)$$

gives (33). □

Corollary 6.3 (Generic paired defect). *If $a_{r+1} \neq 0$, then $n_* = r + 1$ and*

$$H - P_{r+1}^{(m, \Phi)} = (-1)^{r+1}a_{r+1}m^{r(r+1)}x^{2r+3} + O(x^{2r+5}), \quad (34)$$

$$\Lambda - \mathcal{L}_r^{(m)}G_N^{(\Phi, \Lambda)} = (-1)^{r+1}a_{r+1}m^{-r(r+1)}\Lambda^{2r+3}N^{-(2r+2)} + O(N^{-(2r+4)}). \quad (35)$$

Thus every missing coefficient $a_{r+1}, a_{r+2}, \dots$ forces a corresponding automatic order jump. In particular, $a_{r+1} = 0$ raises the scale-side order from $2r + 2$ to at least $2r + 4$.

7 Uniqueness and finite exact recovery

The canonical filter p_r always exists, but jet agreement need not characterize it uniquely when some low active modes are absent.

Let

$$\mathcal{A}_r := \{n \in \{1, \dots, r\} : a_n \neq 0\}, \quad k_r := |\mathcal{A}_r|. \quad (36)$$

Theorem 7.1 (Active-mode uniqueness criterion). *Let q be a polynomial of degree at most r , and set $Q = q(A_m)x$. Then*

$$Q - H = O(H^{2r+3}) \quad (37)$$

if and only if

$$q(1) = 1, \quad q(m^{2n}) = 0 \quad (n \in \mathcal{A}_r). \quad (38)$$

The affine space of such polynomials has dimension

$$\boxed{r - k_r}. \quad (39)$$

Consequently, the jet-recovery filter is unique exactly when

$$\boxed{a_1 a_2 \cdots a_r \neq 0}, \quad (40)$$

in which case the unique solution is $q = p_r$.

Proof. From (13) and Lemma 2.1,

$$q(A_m)x = q(1)H + \sum_{n \geq 1} a_n q(m^{2n})H^{2n+1}.$$

Therefore (37) is equivalent to (38). The nodes

$$1, \quad m^{2n} \quad (n \in \mathcal{A}_r)$$

are pairwise distinct. Evaluation at these $k_r + 1$ nodes has full rank on the $(r+1)$ -dimensional space of polynomials of degree at most r . Hence the solution affine space has dimension $(r+1) - (k_r + 1) = r - k_r$. It is zero-dimensional exactly when $k_r = r$, equivalently (40). In that case the unique polynomial has value one at 1 and zeros at $m^2, \dots, m^{2r}$, hence is precisely p_r . $\square$

Corollary 7.2 (Finite exact recovery). *If*

$$a_n = 0 \quad (n \geq r + 1), \quad (41)$$

then

$$\boxed{P_{r+1}^{(m, \Phi)} = H} \quad (42)$$

as analytic germs, and

$$\boxed{\mathcal{L}_r^{(m)} G_N^{(\Phi, \Lambda)} = \Lambda} \quad (43)$$

for all sufficiently large N .

Proof. Under (41), the exact tails in (16) and (28) are empty. $\square$

8 Recursion and the polynomial-map corollary

The canonical filters satisfy

$$p_j(z) = p_{j-1}(z) \frac{m^{2j} - z}{m^{2j} - 1}. \quad (44)$$

Proposition 8.1 (One-step canonical lift). *For $j \geq 1$,*

$$\boxed{P_{j+1}^{(m, \Phi)} = \frac{m^{2j} P_j^{(m, \Phi)} - A_m P_j^{(m, \Phi)}}{m^{2j} - 1}.} \quad (45)$$

Equivalently,

$$\boxed{P_{j+1}^{(m, \Phi)} = P_j^{(m, \Phi)} + \frac{m P_j^{(m, \Phi)} - P_j^{(m, \Phi)} \circ S_m}{m^{2j+1} - m}.} \quad (46)$$

Proof. Equation (45) is (44) with $z = A_m$, applied to x . Using

$$A_m P_j^{(m, \Phi)} = \frac{1}{m} P_j^{(m, \Phi)} \circ S_m$$

and rearranging yields (46). $\square$

For a general analytic germ, S_m and $P_{r+1}^{(m, \Phi)}$ need not be polynomials, so a universal degree statement is not available. It reappears exactly when the multiplication map is polynomial.

Corollary 8.2 (Polynomial multiplication maps). *Assume S_m is a polynomial of exact degree $d \geq 2$. Then*

$$\deg P_{r+1}^{(m,\Phi)} = d^r. \quad (47)$$

Proof. Since $\deg S_m^{sq} = d^q$, the deepest iterate in (19) has degree d^r , while all preceding terms have smaller degree. Its coefficient is

$$\lambda_{r,r} = \frac{(-1)^r}{\prod_{j=1}^r (m^{2j} - 1)} \neq 0,$$

so the highest-degree term cannot cancel. $\square$

For $\Phi(t) = \sin t$ and odd integer m , one has $S_m(x) = \sin(m \arcsin x)$, an odd polynomial of degree m , and (47) reduces to $\deg P_{r+1} = m^r$. More explicitly,

$$S_m(x) = (-1)^{(m-1)/2} T_m(x),$$

where T_m is the Chebyshev polynomial of the first kind. Thus the multiple-angle composition law and the exponential degree growth are classical Chebyshev structure, not novelty claims.

Remark 8.3 (Compositional economy versus polynomial degree). In the sine case, P_{r+1} has degree m^r , whereas the Taylor polynomial of $\arcsin x$ through degree $2r+1$ already has error $O(x^{2r+3})$. The forward-iterate construction is therefore not polynomial-degree efficient. Its structural advantage is different: the stage- r approximant is assembled from r forward compositions of the known map S_m and fixed scalar weights, without direct evaluation of H . For a general analytic germ the admissible neighborhood also depends on the largest forward node $m^r H(x)$; no claim of fixed- x convergence as $r \rightarrow \infty$ is made.

9 Role of oddness and finite smoothness

Remark 9.1 (What oddness does). The spectral identity (4) does not require Φ to be odd. If a general normalized local diffeomorphism has

$$\Phi(H) = H + \sum_{k \geq 2} c_k H^k,$$

then H^k is still an A_m -eigenmode with eigenvalue m^{k-1} , while

$$N\Phi(\Lambda/N) = \Lambda + \sum_{k \geq 2} c_k \Lambda^k N^{-(k-1)}.$$

Oddness is therefore what restricts the active spectrum to $m^2, m^4, m^6, \dots$ and the scale errors to $N^{-2}, N^{-4}, N^{-6}, \dots$. It is essential to the clean even-power Richardson ladder used in this note, but not to the underlying Schröder–Richardson mechanism itself.

Remark 9.2 (Regularity). Analyticity is convenient because it makes (16) and (28) convergent identities. At the purely algebraic level the entire construction is valid in the ring of odd formal power series

$$\Phi(t) \in t + t^3 \mathbb{R}[[t^2]].$$

For a fixed generic stage r , only finitely many Taylor coefficients and sufficiently many derivatives are needed to obtain the leading asymptotic formulas. In particular, an odd C^{2r+5} local diffeomorphism suffices for the generic defect through the stated remainder order, by the standard Taylor formula with remainder applied to the finitely many compositions occurring at that stage.

10 Unified theorem

The preceding results may be compressed as follows.

Theorem 10.1 (Finite forward-iterate Richardson principle). *Let Φ be an odd analytic germ satisfying*

$$\Phi(0) = 0, \quad \Phi'(0) = 1, \quad \Phi(t) = t + \sum_{n \geq 1} a_n t^{2n+1},$$

let $H = \Phi^{-1}$, and let $m > 1$. Define $S_m, A_m, p_r, P_{r+1}, \ell_r, G_N^{(\Phi, \Lambda)}$, and $\mathcal{L}_r^{(m)}$ by (2), (9), (10), (11), (20), and (21). Then:

(a) $A_m(H^{2n+1}) = m^{2n} H^{2n+1}$, and $\ell_r(z) = z^r p_r(z^{-1})$.

(b) *The exact local remainder is*

$$H - P_{r+1}^{(m, \Phi)} = - \sum_{n \geq r+1} a_n p_r(m^{2n}) H^{2n+1}.$$

(c) *The exact finite-stage intertwining is*

$$\mathcal{L}_r^{(m)} G_N^{(\Phi, \Lambda)} = m^r N P_{r+1}^{(m, \Phi)} \left(\Phi \left(\frac{\Lambda}{m^r N} \right) \right).$$

(d) *The exact scale-side remainder is*

$$\Lambda - \mathcal{L}_r^{(m)} G_N^{(\Phi, \Lambda)} = - \sum_{n \geq r+1} a_n p_r(m^{2n}) m^{-2rn} \Lambda^{2n+1} N^{-2n}.$$

(e) *If $n_\star = \min\{n \geq r+1 : a_n \neq 0\}$, then the first surviving local and scale coefficients are given by (32) and (33). In particular, missing active coefficients produce automatic superconvergence.*

(f) *Jet recovery through order $2r+1$ uniquely determines the canonical degree- r filter if and only if $a_1 \cdots a_r \neq 0$. In general the family of admissible filters has affine dimension $r - k_r$.*

(g) *If $a_n = 0$ for all $n \geq r+1$, then both the local and sequence recovery are exact after finitely many stages.*

11 Three immediate specializations

11.1 Sine/arcsine family

Take $\Phi(t) = \sin t$. Then $H(x) = \arcsin x$ and

$$a_n = \frac{(-1)^n}{(2n+1)!}.$$

Every active coefficient is nonzero. Hence the canonical filter is unique at every stage, and Corollary 6.3 gives

$$\arcsin x - P_{r+1}^{(m, \sin)}(x) = \frac{m^{r(r+1)}}{(2r+3)!} x^{2r+3} + O(x^{2r+5}),$$

while

$$\Lambda - \mathcal{L}_r^{(m)} \left[N \sin \left(\frac{\Lambda}{N} \right) \right] = \frac{\Lambda^{2r+3}}{(2r+3)! m^{r(r+1)}} N^{-(2r+2)} + O(N^{-(2r+4)}).$$

If m is an odd integer, then $S_m(x) = \sin(m \arcsin x)$ is a polynomial of degree m , so Corollary 8.2 gives

$$\deg P_{r+1}^{(m, \sin)} = m^r.$$

For such odd integer m , the multiple-angle composition law gives

$$S_m^{sq}(x) = \sin(m^q \arcsin x), \quad x \in [-1, 1].$$

Hence the finite-stage remainder identity extends from a germ identity to every $x \in [-1, 1]$:

$$\arcsin x - P_{r+1}^{(m, \sin)}(x) = - \sum_{n \geq r+1} \frac{(-1)^n}{(2n+1)!} p_r(m^{2n}) (\arcsin x)^{2n+1}.$$

This global finite-stage identity should not be confused with convergence of $P_{r+1}(x)$ for fixed nonzero x as $r \rightarrow \infty$, which is not asserted.

11.2 Finite exact recovery after one stage

Let

$$\Phi(t) = t + ct^3.$$

Then the only nontrivial active coefficient is $a_1 = c$. For $r = 1$, Theorem 3.1 has an empty tail, so

$$P_2^{(m, \Phi)} = H$$

exactly as germs. Likewise

$$G_N^{(\Phi, \Lambda)} = \Lambda + c\Lambda^3 N^{-2},$$

and the one-stage scale filter gives

$$\frac{m^2 G_{mN}^{(\Phi, \Lambda)} - G_N^{(\Phi, \Lambda)}}{m^2 - 1} = \Lambda$$

exactly.

11.3 A one-stage superconvergent gap

Suppose

$$\Phi(t) = t + a_1 t^3 + a_3 t^7 + O(t^9), \quad a_3 \neq 0,$$

so the coefficient a_2 is absent. With $r = 1$, the first unfiltered active index is $n_\star = 3$. Since

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix}_{m^2} = 1 + m^2,$$

Theorem 6.2 gives

$$H - P_2^{(m, \Phi)} = a_3 m^2 (1 + m^2) x^7 + O(x^9),$$

and

$$\Lambda - \mathcal{L}_1^{(m)} G_N^{(\Phi, \Lambda)} = a_3 (1 + m^2) m^{-4} \Lambda^7 N^{-6} + O(N^{-8}).$$

Thus the nominal one-stage scale order N^{-4} jumps automatically to N^{-6} because the first unfiltered mode is missing.

12 Conclusion

The central identity of this note is elementary once the appropriate Richardson family is exposed:

$$T_y(m^q) = m^{-q} S_m^{\circ q}(x), \quad y = H(x).$$

It identifies a geometric Richardson extrapolant in the Schröder coordinate with a fixed-weight linear combination of finite forward iterates of the original repelling map. The reciprocal relation

$$\ell_r(z) = z^r p_r(z^{-1})$$

then connects this forward realization to the corresponding shrinking-scale filter.

The extrapolation mechanism, geometric-node weights, skipped-mode order jumps, and Schröder–Koenigs diagonalization are classical. The point of the present formulation is instead to make the finite forward-iterate realization explicit and to record the exact paired tails

$$H - P_{r+1}^{(m, \Phi)} = - \sum_{n \geq r+1} a_n p_r(m^{2n}) H^{2n+1}$$

and

$$\Lambda - \mathcal{L}_r^{(m)} G_N^{(\Phi, \Lambda)} = - \sum_{n \geq r+1} a_n p_r(m^{2n}) m^{-2rn} \Lambda^{2n+1} N^{-2n}.$$

In the generic case the first formula gives

$$H - P_{r+1}^{(m, \Phi)} = (-1)^{r+1} a_{r+1} m^{r(r+1)} x^{2r+3} + O(x^{2r+5}),$$

while the reciprocal scale side carries the corresponding $m^{-r(r+1)}$ factor.

For the sine family with odd integer m , the forward maps are Chebyshev multiple-angle polynomials and the finite-stage remainder is global on $[-1, 1]$, but the construction is not degree-efficient relative to Taylor approximation. Its value is compositional: the approximant uses only evaluations of the known forward map and fixed weights. This distinction, together with the paired exact remainders, is the appropriate scope of the present note.

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